

JPH06178281A: Bit error detection method

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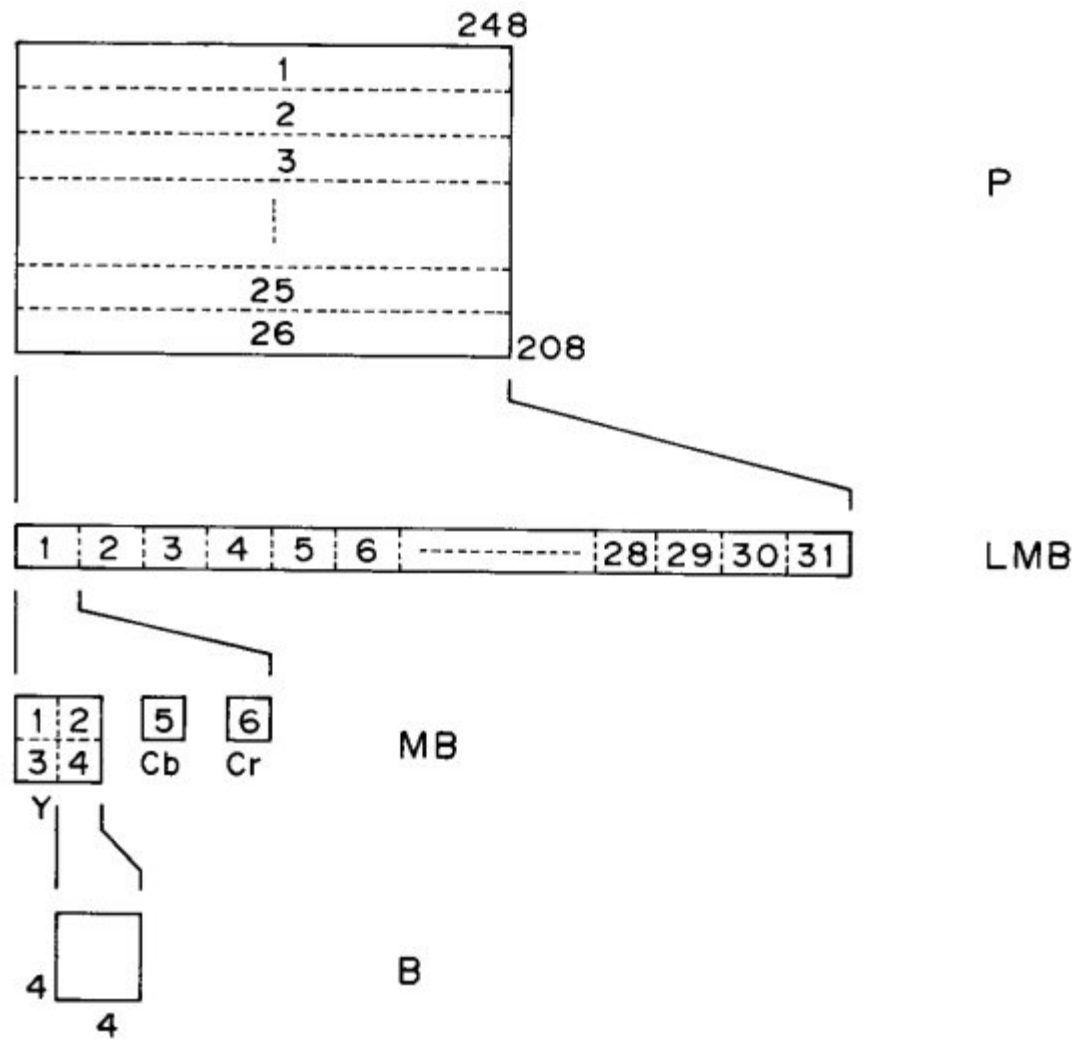
Claim(s)

1. A bit error detection method comprising: an image data pre-processing step of dividing an image signal into a plurality of blocks; a variable-length coding step of adding an EOB code to each of the blocks to obtain compressed image data having a hierarchical structure; and a compressed data reproduction step of counting the number of the EOB codes and determining a bit error according to whether or not the number of the EOB codes is a predetermined number.
2. The bit error detection method according to claim 1, wherein the compressed data includes layers of a block, a macroblock, a line of macroblocks, and a picture, and in the compressed data reproduction step, a bit error of the compressed image data is detected based on whether or not a relationship between a line number of a macroblock in header attribute information of a line of macroblocks layer and the number of counted EOB codes is correct.

Detailed description of the invention

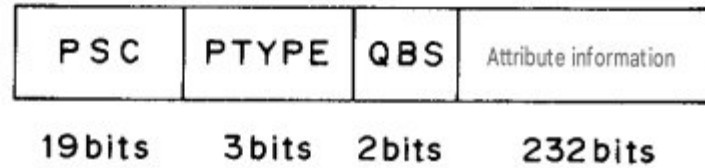
1. The present invention relates to a bit error detecting method for easily detecting a bit error when compressed image data by high efficiency coding is reproduced.
2. Conventionally, there is known an image compression coding method in which compressed image data has a hierarchical structure composed of four layers. The four layers are blocks, macroblocks, lines of macroblocks, and pictures. Each block contains compressed image data and has a hierarchical structure in consideration of two dimensional orthogonal transform in units of blocks.
3. However, in such a conventional image compression coding system, since compressed image data is contained in each block, if an error occurs even in one bit, the error is propagated to the subsequent blocks, so that the image quality is greatly deteriorated, but it is difficult to detect the bit error.
4. It is an object of the present invention to provide a bit error detection method capable of easily detecting a bit error at the time of reproduction of compressed image data using a hierarchical structure.
5. According to another aspect of the present invention, there is provided a compressed data reproducing method comprising the steps of:
 - a. dividing an image signal into a plurality of blocks
 - b. adding an EOB code to each of the blocks to obtain compressed image data having a hierarchical structure
 - c. and counting the number of the EOB codes and determining a bit error according to whether or not the number of the EOB codes is a predetermined number.
6. Further, in the present invention, it is preferable that the compressed data is composed of layers of a block, a macroblock, a line of macroblocks, and a picture, and in the compressed data reproducing step, a bit error of the compressed image data is detected depending on whether or not a relationship between a line number of a macroblock in header attribute information of a line of macroblocks layer and the number of counted EOB codes is correct.
7. In the present invention, an EOB code is always added to each block of compressed image data having a hierarchical structure, the number of EOB codes is counted for each line-of-macroblock layer in a reproduction system, and a bit error is detected by checking the relationship between the number of blocks and the number of macroblock lines. Therefore, according to the present invention, although the compression ratio is somewhat lowered, the accuracy of error detection is improved, and as a result, the image quality is improved.
8. Hereinafter, an embodiment of the present invention will be described in detail with reference to the drawings.

[Fig. 1]



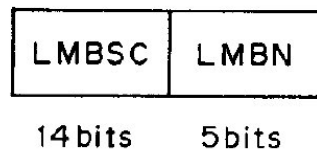
- a. FIG. 1 shows an example of the hierarchical structure of compressed image data used in an embodiment of the present invention. The compressed image data has a hierarchical structure including a block (B), a macroblock (MB), a line of macroblocks (LMB), and a picture (P). One block consists of 4 pixels $\times$ 4 pixels, one macroblock consists of 6 blocks including luminance and chrominance information, one line of macroblocks consists of 31 macroblocks, and one picture consists of 26 lines of macroblocks. As a result, one picture has 4836 blocks and represents one screen of 208 pixels $\times$ 248 pixels.

[Fig. 2]



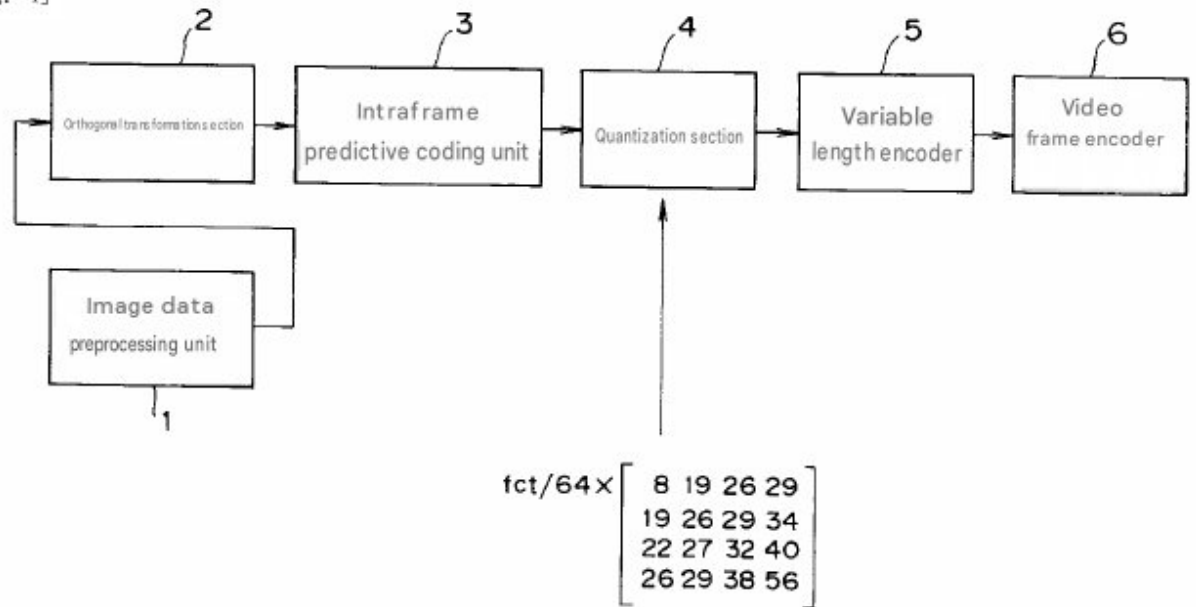
- b. One header attribute information as shown in FIG. 2 is attached to each picture layer. In FIG. 2, PSC is a picture start code, PTYPE is a type information, QBS is a quantization index bit shift, and attribute information includes various factor values, Y / C quantization coefficient table values, and the like.

[Fig. 3]



- c. In addition, one piece of header attribute information as shown in FIG. 3 is added to each line-of-macroblock layer. In FIG. 3, LMBSC is a line-of-macroblock start code, and LMBN is a line-of-macroblock number in which 1 to 26 indicating the macroblock line number are entered. There is no header attribute information in the macroblock and block layers.

[Fig. 4]

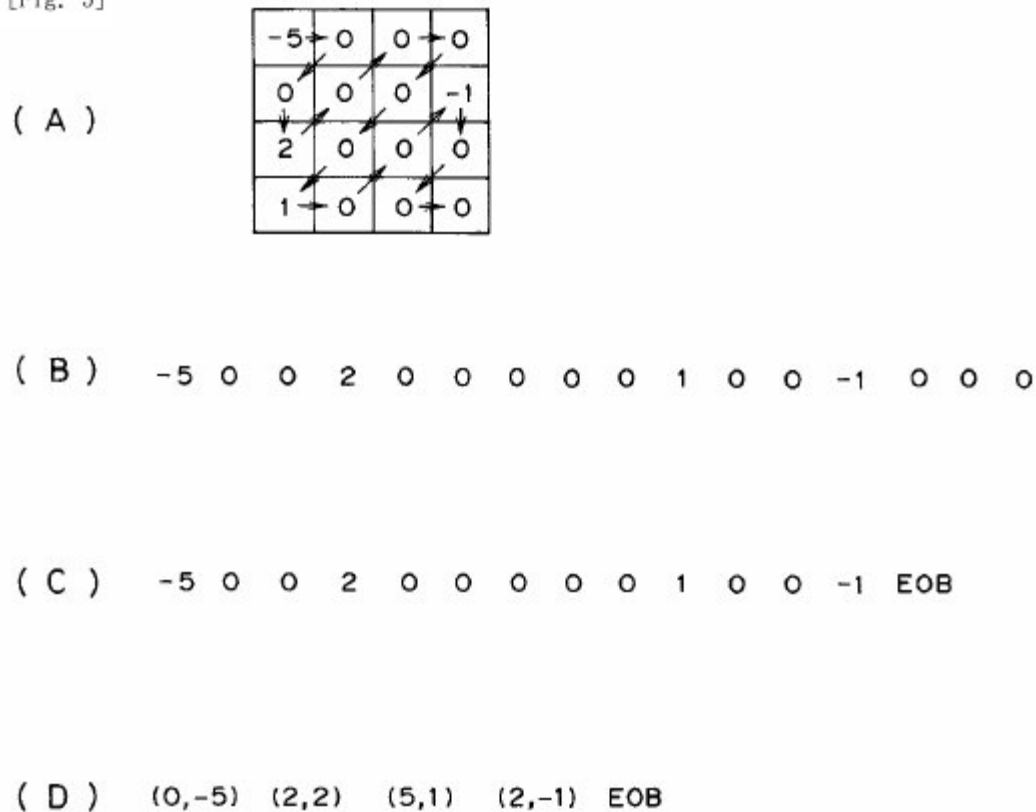


- d. FIG. 4 shows functional blocks of an image compression algorithm executed by the image compression encoder according to the embodiment of the present invention. The contents of FIG. 4 will be described along the flow of processing. In order to improve the compression effect, the image data preprocessing unit 1 digitizes 512 times 480 pixels of one screen of the input digital image data into 248 times 208 pixels, and blocks the pixels into small areas to obtain image data having a hierarchical structure as shown in FIG. 1.
- e. Next, in the orthogonal transform coding unit 2, transform to the frequency domain is performed by two dimensional Hadamard transform accompanied by coordinate axis rotation or two dimensional discrete cosine transform (DCT) or the like for each two dimensional image block of 4×4 pixels, and transform coefficients of respective frequency components arranged in descending order of power and expressed by linear combination are coded. At this time, a high-frequency component having a small power component is deleted in order to realize high compression coding.
- f. Next, the intra-frame predictive coding unit 3 performs intra-frame predictive coding of the DC component of each of Y (luminance), Cb (blue color difference), and Cr (red color difference) on the orthogonal transform output. That is, the data of the current pixel of interest is predicted from the data of the immediately preceding macroblock, and only the difference between the predicted value and the value of the input signal is encoded. In other words, the inter-pixel distance is encoded with a pixel for which prediction has failed as a significant pixel. In this case, the data of the macro block at the head of the preceding screen is distributed to the first macro block of one screen.
- g. Next, in the quantization unit 4, quantization (non-linear quantization) of a different quantization width for each frequency component is performed on the

predictive coding output in order to match the visual system. That is, adaptive bit allocation is performed in such a manner that a high frequency portion having a small energy component and a low visual sensitivity is decreased by increasing the step width, and a low frequency portion having a high visual sensitivity is decreased by decreasing the step width.

- h. Next, the variable-length coding unit 5 converts the two dimensional quantized coefficient data of the block, which is the quantized output of the orthogonal transform coefficient, into a Huffman code while performing zigzag scanning from a low order (low-frequency component) to a high order (high-frequency component).
- i. That is, in general, as the order increases, the number of terms that are actually regarded as "0" among the transformed components increases. This leads to more efficient transmission. Therefore, the coding is performed in accordance with the scanning order, and the coding is performed up to the last coefficient whose quantized coefficient is not "0", and an EOB (End of Block) code indicating the end of the aborted scanning is added to the remaining coefficients.
- j. The one dimensional data obtained by the zigzag scanning includes several continuous "0" s and non-zero data, and two dimensional Huffman coding for generating one code is performed on a two dimensional code (symbol) of a combination of the number of continuous "0" s (run length) and non-zero data (referred to as a significant conversion coefficient). In Huffman coding, a shorter code is given to a symbol having a higher occurrence probability, thereby minimizing the average code length and increasing the transmission efficiency. Since the amount of data after Huffman coding varies depending on the picture, the data is once stored in a buffer memory and then read out at a constant speed.
- k. Finally, in the video frame encoding unit 6, the header attribute information as shown in FIGS. 2 and 3 described above is added and formatted into a frame configuration.
- l. The compressed image data thus compressed is recorded on a recording medium such as a floppy disk, or parallel-to-serial converted and transmitted to a remote apparatus via a transmission line.

[Fig. 5]

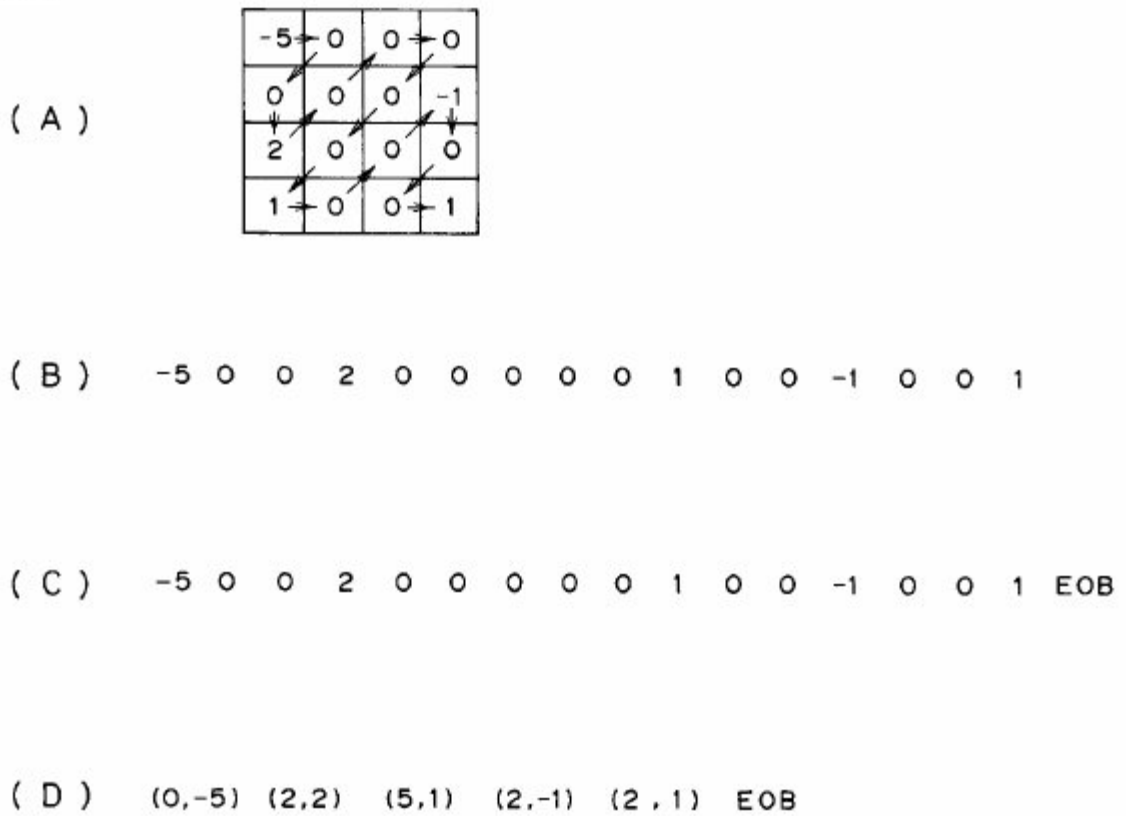


- m. FIG. 5 shows a procedure for converting data of quantized coefficients of one block into Huffman codes in the variable-length coding unit 5 by using specific numerical examples. First, as indicated by an arrow in FIG. 5 a, the quantization coefficient QINDEX in the 4×4 block is scanned in a zigzag manner. The scanning result of the quantization coefficients is shown in FIG. 5 b. Then, as shown in FIG. 5 c, the EOB is inserted immediately after the last non-zero coefficient of the scan, thereby increasing the transmission efficiency of the subsequent zero coefficients.

The quantized coefficients are shown in FIG. 5

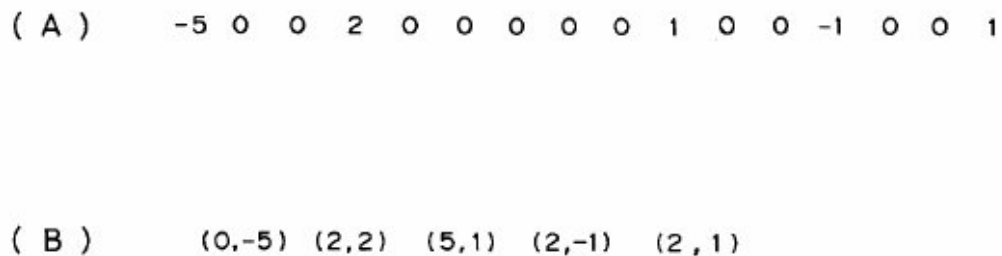
As shown in (D), a two dimensional code (R, L) obtained by combining the run length R of a zero coefficient and the level L of a non-zero coefficient following it is used. In the two dimensional code (R, L), R represents the number of zero components preceding non-zero components by an unsigned integer of 4bits (bit), and L represents the level of non-zero components by a signed integer of 8bits. Further, Huffman coding is applied to the events of the two dimensional code. During Huffman coding, the EOB is also counted as one of the occurrences, and a specific bit code is assigned to it. The Huffman code length is up to the longest 14bits.

[Fig. 6]

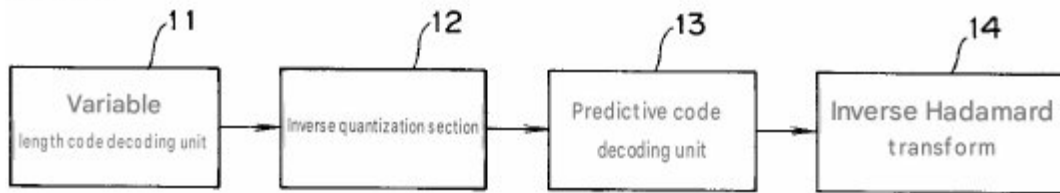


- n. FIG. 6 illustrates the treatment according to the present invention in the case where the last quantized coefficient of "0" is a significant transform coefficient as a result of the zigzag scan described above. The result of the zigzag scanning of the quantized coefficients of the block shown in FIG. 6 a is shown in FIG. 6 b. When the last quantized coefficient is not "0" as described above, the EOB code is not added as shown in FIGS. 7 a and 7 b in the prior art. In the present invention, as will be described later, since the EOB is used as a check code for bit error detection when image data is reproduced, the EOB is inserted even when the last quantized coefficient is not "0", as shown in FIGS. 6 c and 6 d. That is, in the present invention, an EOB (End of Block) code is always inserted for each block.

[Fig. 7]

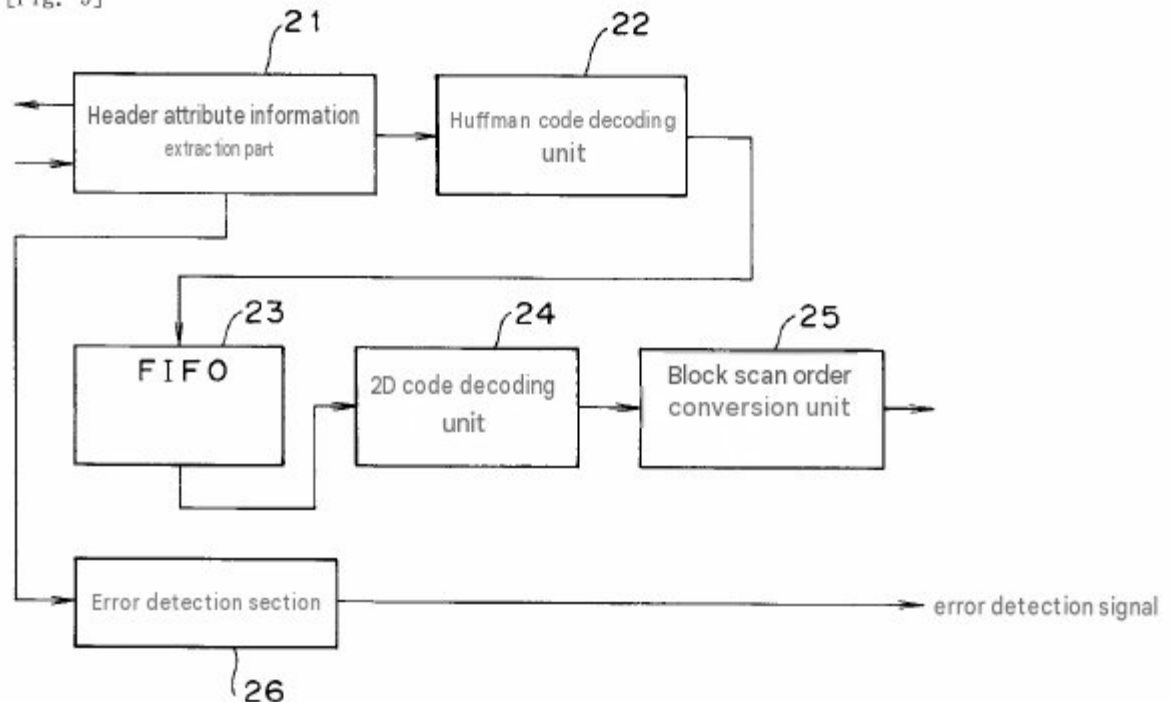


[Fig. 8]



- o. Next, the reproduction processing of the compressed image data in the embodiment of the present invention will be described with reference to FIG. 8. The compressed data recorded on the recording medium or transmitted is temporarily stored in a buffer memory (not shown) and then supplied to the compressed image reproducer shown in FIG. 8. In this compressed image reproducer, the variable-length code decoding unit 11, the inverse quantization unit 12, the predictive code decoding unit 13, and the inverse Hadamard transform unit 14 perform an operation opposite to that of the image compression encoder shown in FIG. 4 to reproduce an image signal before encoding. First, the variable-length code decoding unit 11 converts the Huffman code into quantized coefficient data of one block by performing an operation opposite to the variable-length coding process, as will be described later. Then an inverse quantization section 12 converts the quantization coefficient QINDEX depending on the value of the quantization coefficient QINDEX to realize non-linear quantization. Next, the predictive coding / decoding unit 13 adds the input data and the predicted value to obtain data in a state before predictive coding. Next, the inverse Hadamard transform unit 14 performs a 4. times. 4 two dimensional inverse Hadamard transform on the 1, 0-bit Hadamard index coming from the inverse quantization unit 12 by pipeline processing to obtain actual image data.

[Fig. 9]

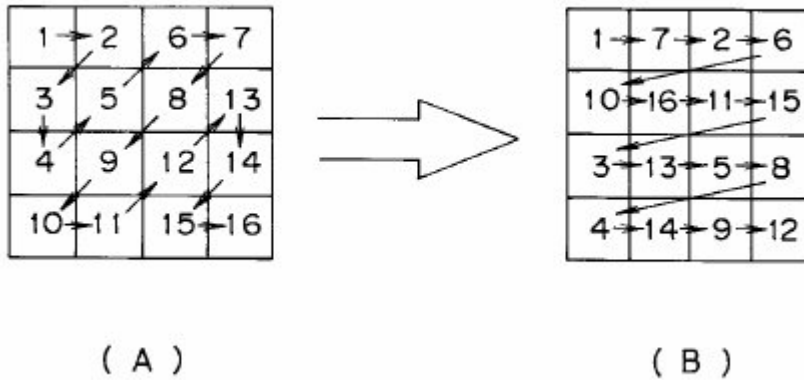


- p. FIG. 9 shows a detailed configuration of the variable-length code decoding unit 11 shown in FIG. 8. First, the header attribute information extraction unit 21 extracts the following header attribute information in the input image encoded data (compressed image data), and performs predetermined processing. That is, a 19-bit PSC (picture start code) in the header attribute information of the picture layer is detected, and the octet counter of the picture header is initialized. The Y and C factors and the Y and C quantization table information in the picture property information are extracted, and the picture property information is output at the timing of a predetermined timing chart in synchronization with the output data of the quantization coefficient QINDX. The 19-bit LMBSC (line of macroblock start code) and LMBN (line of macroblock number) of the header property information of the line of macroblock layer are detected, and the variable-length code decoding unit (VLCD unit) 11 is initialized for each macroblock line.
- q. The decoding of the Huffman codes into the two dimensional codes is performed by referring to the Huffman ROM table in the Huffman code decoding unit 22 in the cycle of 34. 92ns. First, by operating only the Huffman code decoding unit 22 in the 2k cycle without operating the two dimensional code decoding unit 24, 12-bit two dimensional codes (see FIG. 5 (D) or FIG. 6 (D)) are stacked until the FIFO (first in first out memory) 23 of 128 words × 12 bits becomes full in the case of any combination of Huffman codes. When the FIFO23 becomes full, the operation of the Huffman-code decoding unit 22 is temporarily stopped, and the two dimensional-code decoding unit 24 starts to operate in a burst cycle of 139. 68ns. When the two dimensional code decoding unit 24 operates and a free

space is generated in the FIFO23, the Huffman code decoding unit 22 continues to operate until the FIFO23 becomes full.

- r. In the two dimensional code decoding unit 24, the two dimensional code (R, L) obtained by combining the run length: R of the zero coefficient and the level: L of the non-zero coefficient following the zero coefficient is decoded into the scanning result of the quantized coefficient (see FIG. 5 b and FIG. 6 b).

[Fig. 10]



- s. The output of the two dimensional code decoding unit 24 is supplied to the in-block scan order conversion unit 25. In the conversion unit 25, the zigzag scan order in the block is returned to the scan order for each row as shown in FIG. 10, and the scan orders in the row direction and the row direction are converted into the scan order for the high-speed pipeline operation suitable for the inverse Hadamard transformation unit 14, and the converted scan order is output as the quantization coefficient QINDX.
- t. A part of the header attribute information detected by the header attribute information extraction unit 21 is also supplied to the error detection unit 26. The error detector 26 detects an error in the compressed image data by constantly monitoring the relationship between the LMBN (line of macroblock number) and the number of EOB (end of block) in the data.
- u. That is, as described above, the compressed image data used in the present invention has a hierarchical structure, the macroblock line number is entered as header attribute information for each line of macroblock layer, and one EOB code is always inserted for each block regardless of whether or not the last quantized coefficient of the zigzag scanning is zero, as shown in FIGS. 5 and 6. Since the number of blocks in one line of macroblock is constant (31. times. 6 in this embodiment) and one EOB code is always inserted in each block as described above, the number of EOB codes existing in each line of macroblock is constant. However, when a bit error occurs during transmission, the data is shifted, and the relationship between the number of EOB codes and the LMBN changes.
- v. Therefore, the error detection unit 26 of the compressed image reproducer detects the EOB code and counts how many EOBs exist in one line of

macroblock. For example, in the case of this embodiment, the error detection unit 26 detects a bit error of the compressed image data depending on whether or not a line-of-macroblock header comes for every 31×6 EOB codes and the relationship with the line number (LMBN) of the macroblock in the header attribute information is correct.

- w. In this way, when a bit error is detected, an error detection signal indicating the fact is sent from the error detection unit 26 to a control unit (not shown). In response to the error detection signal, the control unit performs a predetermined control operation such as discarding the corresponding screen, receiving the compressed data of the same screen again, or stopping the compression and reproduction process.
 - x. The present invention is not limited to the image compression / decoding processing system described in the above embodiment, and can be applied to various hierarchical structures, orthogonal transform systems, predictive coding systems, quantization systems, and variable-length coding systems.
9. Accordingly, various modifications may be made without departing from the spirit or scope of the general inventive concept as defined by the appended claims and their equivalents. EOB codes are always added to each block of compressed image data having a hierarchical structure, the number of EOB codes is counted for each line-of-macroblock layer in the reproduction system, and bit errors are detected by checking the relationship between the number of blocks and the number of macroblock lines.

10. Brief description of the drawings:

- a. FIG. 1 is a conceptual diagram showing an example of a hierarchical structure of compressed image data used in an embodiment of the present invention.
- b. FIG. 2 is a format diagram showing the structure of header attribute information attached to each picture layer in FIG. 1;
- c. FIG. 3 is a format diagram showing the structure of header attribute information attached to each line of macroblock layer in FIG. 1;
- d. FIG. 4 is a block diagram showing functional blocks of an image compression algorithm executed by the image compression encoder according to the embodiment of the present invention;
- e. FIG. 5 is an explanatory diagram showing a conversion procedure according to the embodiment of the present invention when quantized coefficient data of one block is converted into a Huffman code in the variable length coding unit shown in FIG. 4;
- f. FIG. 6 is an explanatory diagram showing a conversion procedure according to an embodiment of the present invention when the last quantized coefficient is not zero.
- g. FIG. 7 is an explanatory diagram showing a conversion procedure in a conventional example given for comparison with FIG. 6.
- h. FIG. 8 is a block diagram showing a configuration of a compressed image reproducer according to an embodiment of the present invention;
- i. FIG. 9 is a block diagram showing a detailed internal configuration of a variable-length code decoding unit shown in FIG. 8;
- j. FIG. 10 is an explanatory diagram showing a scan sequence conversion operation in the intra-block scan sequence conversion unit of FIG. 9;

11. Explanation of letters or numerals

- a. 1. image data pre-processing unit
- b. 2 orthogonal transform unit
- c. 3. intra-frame prediction coding unit
- d. 4 quantization unit
- e. 5. variable-length coding unit
- f. 6 video frame encoding unit
- g. 11. variable-length code decoding unit
- h. 12 inverse quantization unit
- i. 13. prediction code decoding unit
- j. 14 inverse Hadamard transform unit
- k. 21. header attribute information extraction unit
- l. 22 Huffman code decoding unit
- m. 23. FIFO memory
- n. 24 two dimensional code decoding unit
- o. 25. intra-block scan order conversion unit
- p. 26 error detection unit LMB line of macroblock LMBN line of macroblock number EOB end of block